



API Documentation 2021

Introduction

The FocusEconomics API allows subscribers to always have access to the latest data from FocusEconomics and get the data whenever they desire. The overall domain is api.focus-economics.com and the relevant endpoints can be found in this instruction/documentation. FocusEconomics continuously strives to improve its services by adding new data and features. Any new features and additional data made available is clearly communicated to API clients.

Basic information for developers

- We provide access to our RESTful API as a self-service. Subscribers should be able to access our API service based on the documentation provided here alone.
- FocusEconomics does not officially offer an API setup service for clients, however, should you encounter any issues setting up your API connection we would be happy to assist in any way we can. If the event you need to contact us regarding any technical issues, please write to subscribers@focus-economics.com with as much detail as you can about the problem and we will try to assist you.
- FocusEconomics provides a simple python sample code that downloads our latest data release into a pandas DataFrame
- We will add examples on how to use the API in various programming languages on GitHub and will communicate this to clients once it becomes available
- The data is in a flat JSON format
- The use of https methods is supported

Token based Authentication

The API requires the input of tokens as Authorization headers. There are two types of tokens used:

- Refresh token: this token is provided by FocusEconomics when a client first signs up and is used to request access tokens
- Access token: token used in actual data request. It expires after one hour.

POST /v1/token-exchange:

In order to access the dataset via the API, the Refresh token must be used to obtain an Access token – we call this the token exchange.

Send POST request to <https://api.focus-economics.com/v1/token-exchange> with the headers: {'Authorization' : 'JWT **** Insert Refresh token here****'}

A successful response will be a JSON object with a message and an Access token for subsequent data requests (expires after one hour).

Example using demo Refresh token:

```
Request POST \
--url 'https://api.focus-economics.com/v1/token-exchange' \
--header {'Authorization' : 'jwt
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJfaWQiOiI1ZjUwYmYzYzkyY2VkZDAwNTFlYWVjYjgiLCJlbWFpbCI6ImRlbW
9AZGVtby5kZSIsImNsaWVudHNIY3JldCI6IjZjNjA5Y2FmNzlwZjNhOGQwMmZjNWRIYzVmMzRlMjYyY2ZlU0YzAiL
CjYXQiOiE1OTkxMjcNTZ9.jMqLXN3vB9OwmOI_oKy_VFuUoyA8ziqZ9z-02MufKng'} \
Response: {
  "message": "Accesstoken created",
  "accesstoken": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJfaWQiOiI1ZjUwYmYzYzkyY2VkZDAwNTFlYWVjYjgiLCJl
bWFpbCI6ImRlbW9AZGVtby5kZSIsImNsaWVudHNIY3JldCI6IjZjNjA5Y2FmNzlwZjNhOGQwMmZjNWRIYzVmMzRlMjYyY2ZlU0YzAiL
CjYXQiOiE1OTkxMjcNTZ9.jMqLXN3vB9OwmOI_oKy_VFuUoyA8ziqZ9z-02MufKng"} \
```

Requesting data

With the Access token, data can be requested from the endpoints described below provided that the client has the appropriate subscription. The process is similar to the token authentication except for being a GET request and substituting the Refresh with an Access token.

We currently use five different endpoints for API calls (see below) but to illustrate how to obtain the sample data we will be using the endpoint /clients.

Example GET request using Access token from previous example:

```
Request GET \
--url 'https://api.focus-economics.com/v1/clients' \
--header {'Authorization': 'jwt
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJfaWQiOiI1ZjUwYmYzYzkyY2VkZDAwNTFIYWVjYjgiLCJlbWFpbiI6ImRlbW
9AZGVtby5kZSIsImNsaVVudHNY3JldCI6IjZjA5Y2FmNzlwZjNhOGQwMmZjNWRIYzVmMzRlMjYyY2ZlU0YzAiL
CjYXQiOiE1OTkxMzMzMzNDMslmV4cCI6MTU5OTEzNjk0M30.Yez8n3aRBR4d7Oz3sqCpb7VQ3M3uDNiN18yUq70a
gEM}' \
Response will be a JSON data array with the requested dataset \
```

The GET requests are identical across all the below endpoints. Note that since filtering is not yet supported, the returned dataset will include all the data that the client is subscribed to (from the relevant endpoint of course).

Endpoints

Latest data from FocusEconomics reports - /v1/latest

Response fields:

Entity	Name of country or commodity group
FE_CODE	Indicator code
Indicator	Name of indicator or commodity
Frequency	Can be either annual, quarterly, or monthly
Period	Period in DD-MM-YYYY format. Note that:
	Annual uses the first of the year as the reference so 2021 would be 01-01-2021

	Quarterly uses the first day of the last month of the quarter as reference so Q3 2021 would be 01-09-2021
	Monthly uses first day of month as reference so September 2021 is 01-09-2021
Data_Type	There are 8 data types:
	Consensus
	Count
	Forecast (a forecast that is not part of the survey e.g. population forecasts)
	Historical
	Maximum
	Median
	Minimum
	Panelist
Source	Name of source
FE_Publication	FocusEconomics report year and month
FE_Publication_Date	Date of release of FocusEconomics report
Value	Value of indicator

Ratings - /v1/ratings

Response fields:

Country	Name of country or commodity group
Agency	Name of rating agency
Rating_date	Date of rating
Rating	Rating
Outlook	Outlook

Special client data - /v1/clients

This endpoint is reserved for clients who need custom data made available to them. The response fields are thus agreed upon with the specific client.

Example with python

Below is an example of how to make the API calls for our demo dataset from the endpoint /clients using a simple python script to load the data into a pandas Dataframe. The Refresh token in the code is for demo purposes only.

```
import json
import requests
import pandas

### request access token
turl = 'https://api.focus-economics.com/v1/token-exchange'
theaders = {'Authorization': 'jwt eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJfaWQiOiI1ZjUwYmYzYzkyY2VkZDAwNTFIYWVjYjgiLCJlbWFpbCI6ImRlbW9AZGVtby5kZSIsImNsaWVudHNIY3JldCI6IjZjZjA5Y2FmNzlwZjNhOGQwMmZjNWRIYzVmMzRlMjYyZmZlYzU0YzAiLCJpYXQiOiE1OTkxMjc5NTZ9.jMqLXN3vB9OwmOI_oKy_VFuUoyA8ziqZ9z-02MufKng'}
tr = requests.post(turl, headers=theaders)
tres = tr.json()

### get data from /clients
url = 'https://api.focus-economics.com/v1/clients/'
headers = {'Authorization': 'jwt ' + tres['accesstoken']}
r = requests.get(url, headers=headers)
res = r.json()

df = pandas.DataFrame(res)
print(df)
```